

# ZHENRONG SHEN

Email: zhenrongshen@sjtu.edu.cn | [Homepage](#) | [Google Scholar](#) | [GitHub](#)

## EDUCATION

### [Shanghai Jiao Tong University](#) (SJTU)

Shanghai, China

Ph.D. in Biomedical Engineering; Advisor: [Qian Wang](#)

Sep. 2020 - Jun. 2025

- **Ph.D. Thesis:** Research on Controllable Medical Image Synthesis Methods for Intelligent Disease Diagnosis.

### [Beihang University](#) (BUAA)

Beijing, China

B.Eng. in Biomedical Engineering

Sep. 2016 – Jun. 2020

## HONORS & AWARDS

- [Young Scientist Award](#), MICCAI 2024 Oct. 2024
- Academician Yazhu Chen Scholarship, School of Biomedical Engineering, SJTU Dec. 2023
- 2nd Prize in Student Academic Forum, Medical Imaging Computing Seminar (MICS 2023) Jul. 2023
- 3rd Prize in Student Academic Forum, Medical Imaging Computing Seminar (MICS 2022) Jul. 2022
- 2nd Prize, 17th China Post-graduate Mathematical Contest in Modeling Dec. 2020
- Outstanding Graduate Award, BUAA Jun. 2020
- 1st Prize, National Biomedical Engineering Innovation Design Competition for College Students Jul. 2019
- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM) Apr. 2019
- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM) Apr. 2018

## RESEARCH EXPERIENCE

### Controllable Lesion Data Synthesis

1. **Chest X-ray Lung Nodule Synthesis for Lung Nodule Detection** 2020 – 2022
  - Proposed an inpainting-based lung nodule synthesis network and a synthetic sample selection network to select effective synthetic samples for augmenting lung nodule detection. This work was accepted by *PRCV 2021*.
  - Proposed a lung nodule synthesis framework that disentangles nodule attributes (*i.e.*, shape, size, and texture) and synthesize nodules in a controllable manner, and leveraged such controllability to design a hard example mining strategy for augmenting lung nodule detection. This work was accepted by *Medical Image Analysis*.
2. **Cervical Cytological Image Synthesis for Cervical Abnormality Screening** 2022 – 2024
  - Proposed CellGAN, a conditional GAN that generates cytological images of various cervical cell types including NILM, ASC-US, ASC-H, LSIL, and HSIL cells for augmenting patch-level cervical cell classification. This work was accepted by *MICCAI 2023 (early accept)*.
  - Incorporated CellGAN into a knowledge distillation framework for multi-class cervical cell detection, which facilitates the class-balanced pre-training of a teacher network. This work was accepted by *Neural Networks*.

### Cross-modality Medical Image Synthesis

1. **PET Image Synthesis for Parkinson's Disease (PD) Diagnosis** 2023 – 2025
  - Propose a two-stage framework that synthesizes  $^{11}\text{C}$ -CFT PET images from real  $^{18}\text{F}$ -FDG PET scans for multi-modal PD diagnosis, which was based on the correlation between dopaminergic deficiency in the striatum and increased glucose metabolism in PD patients. This work was accepted by *European Journal of Nuclear Medicine and Molecular Imaging*.
  - Proposed a Metabolism-aware Anomaly Detection (MetaAD) framework, which leverages a cyclic modality translation workflow to identify metabolism anomalies of PD in  $^{18}\text{F}$ -FDG PET scans. This work was early accepted by *MICCAI 2024 (Young Scientist Award)*.

## 2. Whole-body MR-to-CT Synthesis for PET Attenuation Correction

2023 – 2025

- Proposed a whole-body MR-to-CT synthesis framework that integrates structural guidance, spatial alignment, and semantic constraint to enhance synthetic CT image quality, thus facilitating PET attenuation correction in whole-body PET/MR Imaging. This work was accepted by *Medical Image Analysis*.

## Medical Image Quality Enhancement

### 1. MRI Super-resolution for Arbitrary MRI Inter-Slice Spacing Reduction

2022 – 2023

- Proposed HiFi-Diff, a diffusion model for arbitrary MRI inter-slice spacing reduction in the through-plane direction by generating in-between MR slices from adjacent MR slices. This work was accepted by *MLMI 2023*.

## JOURNAL PUBLICATIONS

---

### 1. Cross-Modality PET Image Synthesis for Parkinson's Disease Diagnosis: A Leap from [<sup>18</sup>F]FDG to [<sup>11</sup>C]CFT Using Deep Learning

Zhenrong Shen\*, Jing Wang\*, Haolin Huang, Jiaying Lu, Jingjie Ge, Honglin Xiong, Ping Wu, Zizhao Ju, Huamei Lin, Yuhua Zhu, Yunhao Yang, Fengtao Liu, Yihui Guan, Kaicong Sun, Qian Wang, Chuantao Zuo. *European Journal of Nuclear Medicine and Molecular Imaging*, January 2025.

### 2. Image Synthesis with Disentangled Attributes for Chest X-ray Nodule Augmentation and Detection

Zhenrong Shen, Xi Ouyang, Bin Xiao, Jie-Zhi Cheng, Dinggang Shen, Qian Wang. *Medical Image Analysis*, February 2023.

### 3. Structure-Guided MR-to-CT Synthesis with Spatial and Semantic Alignments for Attenuation Correction of Whole-Body PET/MR Imaging

Jiaxu Zheng\*, Zhenrong Shen\*, Lichi Zhang, Qun Chen. *Medical Image Analysis*, April 2025.

### 4. FLEX-MoCo: Flexible MRI Motion Correction Using Motion Recognition and Adaptive Routing

Feng Li\*, Zhenrong Shen\*, Jiangdong Cai, Rongrong Xie, Zijian Zhou, Han Zhang, Dinggang Shen, Feng Shi, Qian Wang. *Pattern Recognition*, April 2026.

### 5. Distillation of Multi-class Cervical Lesion Cell Detection via Synthesis-aided Pre-training and Patch-level Feature Alignment

Manman Fei, Zhenrong Shen, Zhiyun Song, Xin Wang, Maosong Cao, Linlin Yao, Xiangyu Zhao, Qian Wang, Lichi Zhang. *Neural Networks*, October 2024.

### 6. Segment Anything Model for Medical Image Segmentation: Current Applications and Future Directions

Yichi Zhang, Zhenrong Shen, Rushi Jiao. *Computers in Biology and Medicine (Highly Cited Paper)*, March 2024.

### 7. Unleashing the Potential of Generalist Segmentation Foundation Models for Biomedical Image and Video Analysis

Yichi Zhang, Zhenrong Shen, Lanlan Li, Wenbo Zhang, Le Xue. *iRADIOLOGY*, April 2026.

### 8. UniCAS: A Foundation Model for Cervical Cytology Screening

Haotian Jiang\*, Jiangdong Cai\*, Zhenrong Shen, Mengjie Xu, Manman Fei, Haolin Huang, Xinyu Wang, Rui Bi, Dinggang Shen, Lichi Zhang, Qian Wang. *Cell Reports Medicine*, January 2026.

### 9. Improving Self-Supervised Medical Image Pre-Training by Early Alignment with Human Eye Gaze Information

Sheng Wang\*, Zihao Zhao\*, Zhenrong Shen, Bin Wang, Qian Wang, Dinggang Shen. *IEEE Transactions on Medical Imaging*, January 2025.

10. **Learning Contrast and Content Representations for Synthesizing Magnetic Resonance Image of Arbitrary Contrast**  
Honglin Xiong, Yulin Wang, **Zhenrong Shen**, Kaicong Sun, Yu Fang, Yan Chen, Dinggang Shen, Qian Wang.  
*Medical Image Analysis, May 2025.*
11. **Exploring Multi-Connectivity and Subdivision Functions of Brain Network via Heterogeneous Graph Network for Cognitive Disorder Identification**  
Dongdong Chen, Mengjun Liu, **Zhenrong Shen**, Linlin Yao, Xiangyu Zhao, Zhiyun Song, Haolei Yuan, Qian Wang, Lichi Zhang.  
*IEEE Transactions on Neural Networks and Learning Systems, October 2024.*
12. **AdLER: Adversarial Training with Label Error Rectification for One-Shot Medical Image Segmentation**  
Xiangyu Zhao, Sheng Wang, Zhiyun Song, **Zhenrong Shen**, Linlin Yao, Haolei Yuan, Qian Wang, Lichi Zhang.  
*Expert Systems with Applications, January 2026.*
13. **sTBI-GAN: An Adversarial Learning Approach for Data Synthesis on Traumatic Brain Segmentation**  
Xiangyu Zhao\*, Di Zang\*, Sheng Wang, **Zhenrong Shen**, Kai Xuan, Zeyu Wei, Zhe Wang, Ruizhe Zheng, Xuehai Wu, Zheren Li, Qian Wang, Zengxin Qi, Lichi Zhang.  
*Computerized Medical Imaging and Graphics, March 2024.*
14. **A Deep Learning Framework for Synthesizing Longitudinal Infant Brain MRI During Early Development**  
Yu Fang\*, Honglin Xiong\*, Jiawei Huang, Feihong Liu, **Zhenrong Shen**, Xinyi Cai, Han Zhang, Qian Wang.  
*Radiology: Artificial Intelligence, August 2025.*
15. **Uni-COAL: A Unified Framework for Cross-Modality Synthesis and Super-Resolution of MR Images**  
Zhiyun Song, Zengxin Qi, Xin Wang, Xiangyu Zhao, **Zhenrong Shen**, Sheng Wang, Manman Fei, Zhe Wang, Di Zang, Dongdong Chen, Linlin Yao, Qian Wang, Xuehai Wu, Lichi Zhang.  
*Expert Systems with Applications, December 2024.*
16. **Spatial Attention-based Implicit Neural Representation for Arbitrary Reduction of MRI Slice Spacing**  
Xin Wang\*, Sheng Wang\*, Honglin Xiong, Kai Xuan, Zixu Zhuang, Mengjun Liu, **Zhenrong Shen**, Xiangyu Zhao, Lichi Zhang, Qian Wang.  
*Medical Image Analysis, May 2024.*
17. **DW-Net: A Cascaded Convolutional Neural Network for Apical Four-chamber View Ssegmentation in Fetal Echocardiography**  
Lu Xu, Mingyuan Liu, **Zhenrong Shen**, Hua Wang, Xiaowei Liu, Xin Wang, Siyu Wang, Tiefeng Li, Shaomei Yu, Min Hou, Jianhua Guo, Jicong Zhang, Yihua He.  
*Computerized Medical Imaging and Graphics, May 2020.*

## CONFERENCE PUBLICATIONS

---

1. **CellGAN: Conditional Cervical Cell Synthesis for Augmenting Cytopathological Image Classification**  
**Zhenrong Shen**, Maosong Cao, Sheng Wang, Lichi Zhang, Qian Wang.  
*MICCAI 2023 (early accept), October 2023.*
2. **Nodule Synthesis and Selection for Augmenting Chest X-ray Nodule Detection**  
**Zhenrong Shen**\*, Xi Ouyang\*, Zhuochen Wang, Yiqiang Zhan, Zhong Xue, Qian Wang, Jie-Zhi Cheng, Dinggang Shen.  
*Chinese Conference on Pattern Recognition and Computer Vision (PRCV), October 2021.*
3. **MetaAD: Metabolism-Aware Anomaly Detection for Parkinson's Disease in 3D <sup>18</sup>F-FDG PET**  
Haolin Huang\*, **Zhenrong Shen**\*, Jing Wang\*, Xinyu Wang, Jiaying Lu, Huamei Lin, Jingjie Ge, Chuantao Zuo, Qian Wang.  
*MICCAI 2024 (Young Scientist Award), October 2024.*

4. **Arbitrary Reduction of MRI Inter-slice Spacing Using Hierarchical Feature Conditional Diffusion**  
Xin Wang\*, Zhenrong Shen\*, Zhiyun Song, Sheng Wang, Mengjun Liu, Lichi Zhang, Kai Xuan, Qian Wang.  
*MICCAI 2023 Workshop on Machine Learning in Medical Imaging (MLMI 2023), October 2023.*
5. **Refining Cervical Cell Classification with Cytological Knowledge and Optimal Attribute Descriptor Matching**  
Manman Fei, Zhenrong Shen, Mengjun Liu, Zhiyun Song, Yusong Sun, Xu Han, Zelin Liu, Haotian Jiang, Lu Bai, Qian Wang, Lichi Zhang.  
*MICCAI 2025, September 2025.*
6. **MeLo: Low-rank Adaptation Is Better Than Fine-tuning for Medical Image Diagnosis**  
Yitao Zhu, Zhenrong Shen, Zihao Zhao, Sheng Wang, Xin Wang, Xiangyu Zhao, Dinggang Shen, Qian Wang.  
*ISBI 2024 (oral), May 2024.*
7. **One-Shot Traumatic Brain Segmentation with Adversarial Training and Uncertainty Rectification**  
Xiangyu Zhao, Zhenrong Shen, Dongdong Chen, Sheng Wang, Zixu Zhuang, Qian Wang, Lichi Zhang.  
*MICCAI 2023 (early accept), October 2023.*
8. **Consistent 3D Medical Image Inpainting via Implicit Representation and Multi-View Sampling**  
Yuqi Hu, Zhenrong Shen, Zhiyun Song, Dongdong Chen, Liao Wang, Lichi Zhang.  
*ICICSP 2024, September 2024.*
9. **Weakly Semi-Supervised Cervical Lesion Cell Detection via Twin-Memory Augmented Multiple Instance Learning**  
Manman Fei, Zhiyun Song, Zhenrong Shen, Mengjun Liu, Qian Wang, Lichi Zhang.  
*MICCAI 2025, September 2025.*
10. **Learnable Subdivision Graph Neural Network for Functional Brain Network Analysis and Interpretable Cognitive Disorder Diagnosis**  
Dongdong Chen, Mengjun Liu, Zhenrong Shen, Xiangyu Zhao, Qian Wang, Lichi Zhang.  
*MICCAI 2023, October 2023.*
11. **Multi-Tracer Uptake Correction for PET-MR via Aligned-Feature Guidance and Multi-scale Pixel-adaptive Routing**  
Aocheng Zhong\*, Haolin Huang\*, Jing Wang\*, Zhenrong Shen, Haiyu Song, Yuhua Zhu, Yang Liu, Junlei Wu, Qian Wang, Chuantao Zuo.  
*MICCAI 2025 (early accept), September 2025.*
12. **Multi-task Screening for Cervical Diseases via Feature Routing and Asymmetric Distillation**  
Haotian Jiang\*, Haolin Huang\*, Jiangdong Cai, Zhenrong Shen, Manman Fei, Mengjie Xu, Xinyu Wang, Lichi Zhang, Qian Wang.  
*MICCAI 2025 (early accept), September 2025.*
13. **RSAD: Region-Specific Anomaly Detection in fMRI for Disease Diagnosis**  
Yusong Sun, Dongdong Chen, Mengjun Liu, Zhenrong Shen, Zhiyun Song, Yuqi Hu, Manman Fei, Xu Han, Zelin Liu, Xingkai Fang, Lu Bai, Lichi Zhang.  
*MICCAI 2025 (early accept), September 2025.*
14. **Self-supervised Learning with Adaptive Graph Structure and Function Representation for Cross-Dataset Brain Disorder Diagnosis**  
Dongdong Chen, Linlin Yao, Mengjun Liu, Zhenrong Shen, Yuqi Hu, Zhiyun Song, Qian Wang, Lichi Zhang.  
*MICCAI 2024, October 2024.*
15. **CAS-Net: Cross-View Aligned Segmentation by Graph Representation of Knees**  
Zixu Zhuang\*, Xin Wang\*, Sheng Wang\*, Zhenrong Shen, Xiangyu Zhao, Mengjun Liu, Zhong Xue, Dinggang Shen, Lichi Zhang, Qian Wang.  
*MICCAI 2023, October 2023.*

16. **Robust Cervical Abnormal Cell Detection via Distillation from Local-Scale Consistency Refinement**  
Manman Fei, Xin Zhang, Maosong Cao, Zhenrong Shen, Xiangyu Zhao, Zhiyun Song, Qian Wang, Lichi Zhang.  
*MICCAI 2023, October 2023.*
17. **MITracker: Multi-View Integration for Visual Object Tracking**  
Mengjie Xu\*, Yitao Zhu\*, Haotian Jiang, Jiaming Li, Zhenrong Shen, Sheng Wang, Haolin Huang, Xinyu Wang, Han Zhang, Qing Yang, Qian Wang.  
*CVPR 2025 (highlighted), June 2025.*
18. **Gaze-DETR: Using Expert Gaze to Reduce False Positives in Vulvovaginal Candidiasis Screening**  
Yan Kong\*, Sheng Wang\*, Jiangdong Cai, Zihao Zhao, Zhenrong Shen, Yonghao Li, Manman Fei, Qian Wang.  
*MICCAI 2024 (early accept), October 2023.*
19. **Longitudinal Image Synthesis for Early Brain Development with Age Awareness**  
Yu Fang\*, Honglin Xiong\*, Jiawei Huang, Xin Wang, Zhenrong Shen, Feihong Liu, Xinyi Cai, Han Zhang, Qian Wang.  
*ISBI 2025, April 2025.*
20. **Alias-Free Co-modulated Network for Cross-Modality Synthesis and Super-Resolution of MR Images**  
Zhiyun Song, Xin Wang, Xiangyu Zhao, Sheng Wang, Zhenrong Shen, Zixu Zhuang, Mengjun Liu, Qian Wang, Lichi Zhang.  
*MICCAI 2023 (early accept), October 2023.*

## ACTIVITIES

---

- **Journal Reviewer**  
*Medical Image Analysis*  
*IEEE Transactions on Medical Imaging*  
*IEEE Transactions on Neural Networks and Learning Systems*  
*IEEE Transactions on Multimedia*  
*IEEE Journal of Biomedical and Health Informatics*  
*Information Fusion*  
*Pattern Recognition*  
*Neural Networks*  
*NeuroImage*  
*Computerized Medical Imaging and Graphics*  
*Medical & Biological Engineering & Computing*
- **Conference Reviewer**  
*Medical Image Computing and Computer Assisted Interventions (MICCAI)*  
*IEEE International Symposium on Biomedical Imaging (ISBI)*
- **Teaching Assistant**  
*Computer Vision in Biomedical Engineering (SJTU BME7001) at Spring 2024*